- **What is Data?**

Data is a collection of raw facts and figures about something.

Raw Data → unprocessed facts

Information → Processed Data (organized and meaningful)

Example: Shop Sales

Raw Data (unprocessed):

A company collects daily sales data (for a week):

[120, 150, 100, 130, 170, 90, 110]

Information (Processed and meaningful):

- Average sales per day: 124 units
- Highest sales = 170 units (on Day 5)
- Lowest sales = 90 units (on Day 6)
- Trend: Sales dropped on weekends

- ***Types of Data:***

1. Qualitative Data:

Data that describes qualities, categories or names.

For example:

Colors of cars → Read, Blue, Black

Gender → Male, Female

Fruit Names → Apple, Banana, Mango

2. Quantitative Data:

Data that can be measured and written in numbers.

For example:

Age → 12 years, 50 years

Height → 160cm, 190 cm

Marks → 80, 70, 65

- **Working With CSV Files:**

A CSV (Comma Separated View) file is file format for storing tabular data (i.e. rows and columns).

Example csv file content.

```
Name, Age, Marks
Faheem, 20, 85
Taha, 22, 86
Raza, 19, 90
```

Python provides multiple ways for working with csv files.

1. Using built-in *csv module*.
2. Using *pandas library.* (a powerful tool for data analysis)

For using csv module, we need the import the module at the start of program.

import csv

- *Reading from CSV File:*

csv.reader() reads each row of a csv file into a list of strings.

```
import csv

# open and read csv file
with open("students_50_records.csv", mode="r") as file:
    reader = csv.reader(file)
    for row in reader:
        print(row)
```

- ***Writing to CSV file:***

o Write multiple rows

```
import csv

data = [
    ["Name", "Age", "Marks"],
    ["Ali", 20, 85],
    ["Sara", 21, 90],
]

with open("sample csv file.csv", mode= “a” , newline="") as file:
    writer = csv.writer(file)
    writer.writerows(data)
```

o Write single row

```
with open("sample csv file.csv", mode="a", newline="") as file:
    writer = csv.writer(file)
    writer.writerow(["Faheem", 24, 90])
```

- ***Reading CSV as Dictionary:***

Easier to access data by column names.

```
with open("sample csv file.csv", mode="r") as file:
    reader = csv.DictReader(file)
    for row in reader:
        print(row)
```

- ***Write CSV from list of Dictionaries:***

```
data = [
    {"Name": "Ali", "Age": 20, "Marks": 85},
    {"Name": "Sana", "Age": 21, "Marks": 90},
]

with open("students.csv", mode="w", newline="") as file:
    fieldnames = ["Name", "Age", "Marks"]
    writer = csv.DictWriter(file, fieldnames=fieldnames)
    writer.writeheader()  # Write column names
    writer.writerows(data)
```